6.4.1 The genetic code

During translation, a cellular organelle called ribosome "reads" the information in a messenger RNA (mRNA) and uses it to build a polypeptide or chain of amino acids. In an mRNA, three consecutive nucleotides are bases known as codons provide the instructions for building a polypeptide. Each codon codes for a single aminoacid. There are totally 61 codons which codes for amino acids and there are three more codons UAA, UAG and UGA which are known as stop codons as they are signals for termination of translation. The codon AUG codes for the aminoacid methionine and it also acts as a start codon. The genetic code is universal i.e, baring a few rare instances it is followed by all organisms. Similalry the genetic code is degenerate, meaning that one codon can only represent one aminoacid but one aminoacid can be represented by more than one codon. For example the codon UUU represents only phenylalanine. However phenylalanine can be represented by two codons UUU and UUC.

Second letter

First Letter	U	C	A	G	Third Letter
U	UUU, UUC: Phe; UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr; **UAA STOP**; **UAG STOP**	UGU, UGC: Cys; **UGA STOP**; UGG: Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAU: His; CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile; AUG: Met	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn; AAA, AAG: Lys	AGU, AGC: Ser; AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp; GAA, GAG: Glu	GGU, GGC, GGA, GGG: Gly	U C A G

Figure 12 The universal genetic code

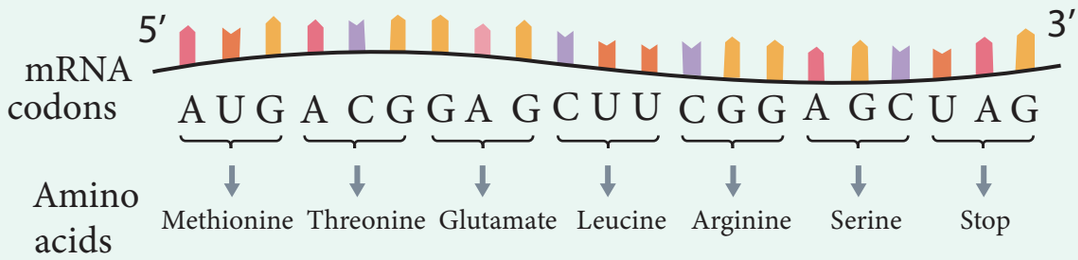


Figure 13 Representational decoding of codons in a sample mRNA to corresponding amino acid sequence.

6.4.2 Overview of translation

How is an mRNA molecule "read" to make a polypeptide? Two important components act as a machinery for this which are the tRNAs and ribosomes. As we have seen already

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